

Submission Worksheet

Submission Data

Course: IT114-450-M2025

Assignment: IT114 Milestone 2 - Trivia

Student: Preet S. (ps55)

Status: Submitted | **Worksheet Progress:** 100%

Potential Grade: 10.00/10.00 (100.00%)

Received Grade: 0.00/10.00 (0.00%)

Started: 7/23/2025 2:05:34 PM

Updated: 8/4/2025 1:30:22 PM

Grading Link: <https://learn.ethereallab.app/assignment/v3/IT114-450-M2025/it114-milestone-2-trivia/grading/ps55>

View Link: <https://learn.ethereallab.app/assignment/v3/IT114-450-M2025/it114-milestone-2-trivia/view/ps55>

Instructions

1. Refer to Milestone2 of [Trivia](#)
 1. Complete the features
 2. Ensure all code snippets include your ucid, date, and a brief description of what the code does
 3. Switch to the Milestone2 branch
 1. `git checkout Milestone2`
 2. `git pull origin Milestone2`
 4. Fill out the below worksheet as you test/demo with 3+ clients in the same session
 5. Once finished, click "Submit and Export"
 6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
 1. `git add .`
 2. ``git commit -m "adding PDF"`
 3. `git push origin Milestone2`
 4. On Github merge the pull request from Milestone2 to main
 7. Upload the same PDF to Canvas
 8. Sync Local
 1. `git checkout main`
 2. `git pull origin main`
- Complete each section and task sequentially.
 - Review the details and validation criteria for each task.
 - Ensure subtasks are completed before the parent task.

Section #1: (1 pt.) Payloads

Progress: 100%

≡ Task #1 (1 pt.) - Show Payload classes and subclasses

Progress: 100%

Details:

- Reqs from the document
 - Provided Payload for applicable items that only need client id, message, and type
 - QAPayload for sending the question and answer options to players (not the correct answer)
 - PointsPayload for syncing points of players
 - Each payload will be presented by debug output (i.e, properly override the toString() method like the lesson examples)

Part 1:

Progress: 100%

Details:

- Show the code related to your payloads (Payload, QAPayload, PointsPayload, and any new ones added)
- Each payload should have an overridden toString() method showing its internal data

```
// Debug-friendly string representation
@Override
public String toString() {
    return String.format(
        format:"[QAPayload] Category: %s | Question: %s | Options: %s",
        category, question, options
    );
}
```

QApayload

```
// Debug-friendly string representation
@Override
public String toString() {
    return String.format(
        format:"[QAPayload] Category: %s | Question: %s | Options: %s",
        category, question, options
    );
}
```

Payload

```
// Debug-friendly string output
@Override
public String toString() {
    return "[PointsPayload] Player Scores: " + playerScores;
}
```


code of both methods

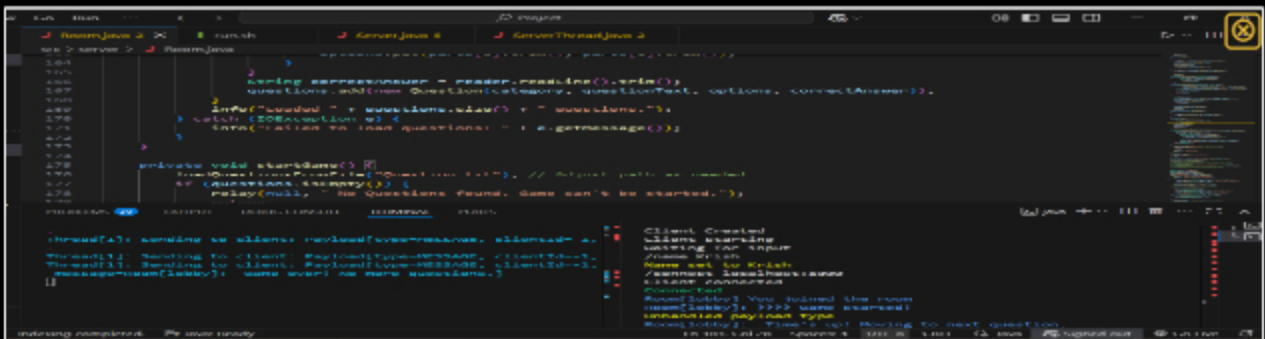
Details:

- Show the snippet of `onSessionStart()`

```
private void startGame() {
    loadQuestionsFromFile("Question.txt"); // Adjust path as needed
    if (questions.isEmpty()) {
        relay(null, "No Questions found. Game can't be started.");
        return;
    }
    gameStarted = true;
    relay(null, "Game started!");
    nextRound();
}

// In Room.java
private void nextRound() {
    if (questions.isEmpty()) {
        relay(null, "Game over! No more questions.");
        gameStarted = false;
        return;
    }
}
```

code



client side



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⇒ Part 2:

Progress: 100%

Details:

- Briefly explain the logic that occurs here (i.e., setting up initial session state for your project) and next lifecycle trigger

Your Response:

The game room loads all trivia questions from a file into memory at the start of the session. It resets player scores and prepares the internal game state for a fresh round. Then, it sends the first question to all players using a `QAPayLoad`. This begins the first round of the game. The next step is to wait for player answers and process them.



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≡ Task #3 (0.80 pts.) - GameRoom Round Start

Progress: 100%

Details:

- Reqs from Document
 - GameRoom randomly chooses and removes a question from the in-memory question list
 - Initialize remaining Players' choices to null (not set)
 - Send the question and answer options to all Clients (without sharing which answer is correct)
 - Set Phase to "choosing"
 - GameRoom round timer begins

Part 1:

Progress: 100%

Details:

- Show the snippet of `onRoundStart()`

[illegible]

```

unhandled payload type
A
KRISHNA: A
Roong Loblay: Time's up! Moving to next question.
unhandled payload type
A
KRISHNA: C
Roong Loblay: Time's up! Moving to next question.
unhandled payload type
Roong Loblay: Time's up! Moving to next question.
unhandled payload type
A
KRISHNA: B
Roong Loblay: Time's up! Moving to next question.
unhandled payload type
KRISHNA: C
Roong Loblay: Time's up! Moving to next question.
unhandled payload type
KRISHNA: C
KRISHNA: C
Roong Loblay: Time's up! Moving to next question.
Roong Loblay: Time's up! Moving to next question.
unhandled payload type
unhandled payload type
A
KRISHNA: B
Roong Loblay: Time's up! Moving to next question.

```

clientside

```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000

```

code

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⇒ Part 2:

Progress: 100%

Details:

- Briefly explain the logic that occurs here (i.e., setting up the round for your project)

Your Response:

At the start of each round, the game picks a random trivia question and sends it to all players without revealing the correct answer. Everyone's previous choices are cleared, so it's a fresh start. The game then enters the "choosing" phase and starts a countdown timer, giving players a limited time to answer.



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≡ Task #4 (0.80 pts.) - GameRoom Round End

Progress: 100%

Details:

- Reqs from Document
 - **Condition 1:** Round ends when all Players locked-in an answer
 - **Condition 2:** Round ends when the Turn timer expires
 - Award points to Players with diminishing returns
 - I.e., Player to correctly answer first gets 10, next gets 7, etc (points are per your choosing as long as it diminishes down to 1)
 - Hint: it's best to have this logic based on the number of Players as a formula rather than being hard-coded
 - Sync the new points for each Player to all Clients
 - Send the in-progress scoreboard to all clients sorted by highest points to lowest
 - Trigger Round Start if end condition not met; otherwise, trigger session end

📁 Part 1:

Progress: 100%

Details:

- Show the snippet of `onRoundEnd()`

```
onRoundEnd() {  
    // Round ends when all Players locked-in an answer  
    // Round ends when the Turn timer expires  
    // Award points to Players with diminishing returns  
    // Sync the new points for each Player to all Clients  
    // Send the in-progress scoreboard to all clients sorted by highest points to lowest  
    // Trigger Round Start if end condition not met; otherwise, trigger session end  
}
```

Terminal

```
onRoundEnd() {  
    // Round ends when all Players locked-in an answer  
    // Round ends when the Turn timer expires  
    // Award points to Players with diminishing returns  
    // Sync the new points for each Player to all Clients  
    // Send the in-progress scoreboard to all clients sorted by highest points to lowest  
    // Trigger Round Start if end condition not met; otherwise, trigger session end  
}
```

Details:

- Show the snippet of `onSessionEnd()`

CODE

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⇒ Part 2:

Progress: 100%

Details:

- Briefly explain the logic that occurs here (i.e., cleanup/reset, next lifecycle events)

Your Response:

At the end of the session, once a predefined number of rounds have been completed—the game wraps up by finalizing and displaying the results. The server calculates the final scoreboard by sorting players based on their total points, then sends it to all clients along with a "Game Over" message. Importantly, players are not kicked out or sent back to the lobby. Instead, their game-related data (like scores and answers) is cleared on both the server and client side to prepare for a future session. To begin a new game, everyone will need to go through a new "ready check" phase, ensuring all players are set before starting again.

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Section #3: (4 pts.) Gameroom User Action And State

Progress: 100%

Task #1 (1.33 pts.) - Question Class

Progress: 100%

Details:

- Reqs from document
 - Will have question text

-
- ```

1 #include <iostream>
2 using namespace std;
3
4 int main() {
5 cout << "Hello, World!" << endl;
6 cout << "Hello, C++!" << endl;
7 return 0;
8 }

```

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## Progress: 100%

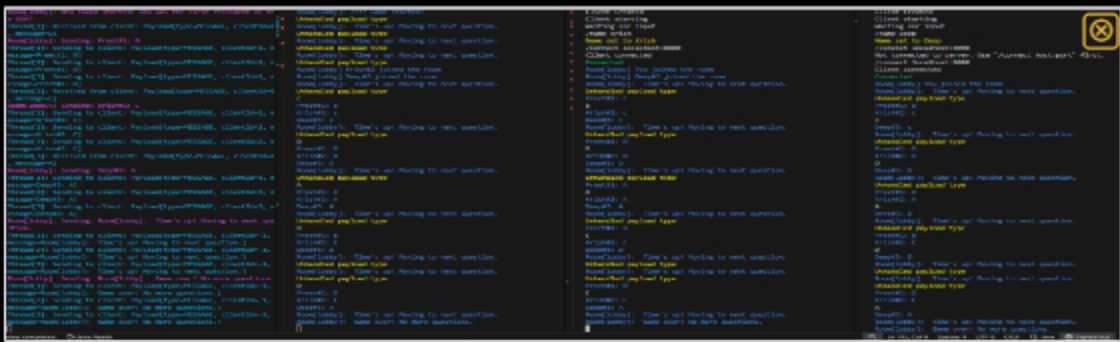
- Reqs from document
  - Command: /answer <choice> (likely A, B, C, or D)
    - Players can make one selection per round (can't change their answer)
    - The choice will be sent to the server-side
      - If the choice is correct, the Player is recorded in the round's "correct" list in the order they correctly answered
    - GameRoom syncs that the Player locked in their answer (it doesn't share what they chose)
      - A message will be relayed saying that "X picked their answer"
    - If all Players have n answer the round ends

## Progress: 100%

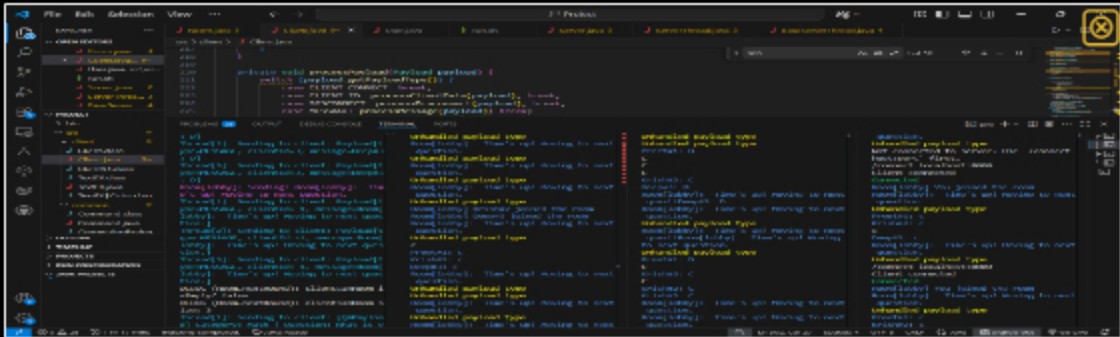
- Show the code snippets of the following, and clearly caption each screenshot
- Show the Client processing of this command (process client command)
- Show the ServerThread processing of this command (process method)
- Show the GameRoom handling of this command (handle method)
- Show the sending/syncing of the results of this command to users (send/sync method)
- Show the ServerThread receiving this data (send method)
- Show the Client receiving this data (process method)







server and client



Q&A

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## Section #4: ( 1 pt.) Misc

Progress: 100%

### Task #1 ( 0.33 pts.) - Github Details

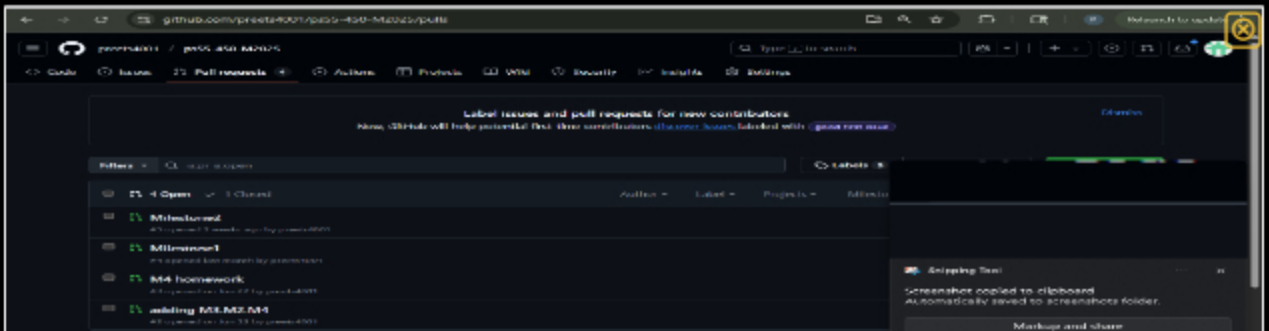
Progress: 100%

#### Part 1:

Progress: 100%

#### Details:

From the Commits tab of the Pull Request screenshot the commit history



commit history



|               |                                          |               |            |
|---------------|------------------------------------------|---------------|------------|
| 2 hrs 55 mins | Project/src/server/Room.java             | 5 hrs 47 mins | Milestone2 |
| 50 mins       | Project/Question.txt                     |               |            |
| 36 mins       | Project/src/Client/Client.java           |               |            |
| 26 mins       | Project/src/server/DataServerThread.java |               |            |
| 14 mins       | Project/src/server/ServerThread.java     |               |            |
| 14 mins       | Project/run.sh                           |               |            |
| 12 mins       | Project/src/common/Question.txt          |               |            |
| 7 mins        | Project/src/common/Question.java         |               |            |
| 3 mins        | Project/build.sh                         |               |            |
| 2 mins        | Project/src/server/Server.java           |               |            |
| 59 secs       | Project/src/server/User.java             |               |            |
| 3 secs        | ...exception/DuplicateRoomException.java |               |            |
| 0 secs        | ...exception/RoomNotFoundException.java  |               |            |
| 0 secs        | ...exception/CustomT134Exception.java    |               |            |
| 0 secs        | Project/src/common/RoomAction.java       |               |            |
| 0 secs        | Project/src/common/Payload.java          |               |            |
| 0 secs        | ...ect/src/common/ConnectionPayload.java |               |            |

waketime2



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## Task #3 ( 0.33 pts.) - Reflection

Progress: 100%

### Task #1 ( 0.33 pts.) - What did you learn?

Progress: 100%

#### Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

I learned how to set up and manage the initial game state for a multiplayer trivia game. This involved reading question data from a file, storing it in memory, and triggering the start of a game session once players joined. I also learned how to structure the game logic so that it can manage client connections, broadcast data to multiple users, and prepare for the first round in a synchronized way. This task deepened my understanding of how server-side coordination works in multiplayer games.



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### Task #2 ( 0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

#### Details:

Briefly answer the question (at least a few decent sentences)

Your Response:

The easiest part of the assignment was reading and loading the trivia questions from a file into memory. It involved basic file I/O operations and simple data parsing, which are familiar and straightforward tasks. Once the format of the question file was clear, implementing the logic to store questions in a list for later use was simple and required minimal debugging.



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## ⇒ Task #3 ( 0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

### Details:

Briefly answer the question (at least a few decent sentences)

### Your Response:

The hardest part of the assignment was coordinating the game flow logic on the server, especially managing the timing of rounds, tracking player answers, and ensuring proper synchronization. Handling concurrent client actions like locking in answers, awarding points based on answer order, and ending rounds based on time or all players responding required careful planning and testing. It was challenging to make sure all conditions were handled correctly without causing race conditions or inconsistent game state.



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